

Semester: V
No.

Roll

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Course Name: B Sc/ Microbiology
Course Code: B-305
Branch(s): Microbiology
60

Pages: 2
Time: 120 Min
Max. Marks:

COs: CO-1 & CO-2

Instructions:
1. Attempt all the sections.
2. Assume missing data suitably, if any.
CO1: On completion of this course, the student will be able to illustrate the fundamentals of Recombinant DNA Technology.
CO-2: On completion of this course, the student will be able to learn the synthesis of fundamental tools required in RDT.
KL: Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Attempt all qu

Q.No.	Questions	CO	
1.	a/ Explain contribution of Arber and Smith in RDT	1	
	b/ Discuss about the Plasmid as cloning vector	1	K1
	c/ Write about the definition of gene Cloning and its role in the field of Medicine	1	K2
	d/ What are palindromic sequences, explain with a suitable example.	2	K1
	e/ Explain the role of enzyme DNA polymerase I	2	K2
	f/ Comment on Lac-Z Gene role in Blue/White screening	2	K3

Section-B

Attempt any two questions of this section.
Marks)

(7 x 2 = 14

Q.No.	Questions	CO	
2.	Describe different kinds of plasmid vectors used for gene cloning.	1	
3.	What are the features of a good vector? Describe the feature of one phage vector of <i>Escherichia coli</i> .	1	K1
4.	Describe the contribution of Mullis in recombinant DNA Technology.	1	K1

Attempt any two questions of this section.

(7 x 2 = 14 M

5.	Explain selectable markers used in cloning vectors		
6.	Write a detail note on Restriction enzymes used in Recombinant DNA Technology	2	K3
		2	K2

7.	Define pBR322 vector with suitable diagram	2
----	--	---

Page: 0

Attempt all questions of this section. (Marks)		<u>Section-D</u> (10 x 2 = 20)	
Q.No.	Questions	CO	K
8.	Describe in detail the procedure of gene cloning and its uses	1	K1
	Or		
9.	Explain the requirement of recombinant molecules in Medical, Agriculture and Industrial sectors.	1	K2
	Explain lambda vector in detail. Compare the applications of lambda vector over plasmid vector	2	K3
	Or		
	Expand full form of YAC vector, explain its properties, selection and applications	2	K3